

PRAKTIKUM SISTEM OPERASI

MODUL VI

PROSES SEKUENSIAL DAN KONKUREN



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PROGRAM STUDI S1 SOFTWARE ENGINEERING

FAKULTAS INFORMATIKA

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JURNAL

1. Selain hardware (memory), batasan maksimal proses dapat ditentukan dengan secara software. Pada Linux maksimal proses adalah 4194303 proses (64 bit) dan 32767 proses (32 bit) dapat dilihat melalui perintah `$cat /proc/sys/kernel/pid_max`

- File process.h

```
Activities Terminal Fri 1 May, 17:33
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help

xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$ grep -r "NPROC" ~/xinu/include/process.h
#ifdef NPROC
    8
    ((pid32)(x) >= NPROC) || \
xinu@xinu-develop-end:~$ cat ~/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifdef NPROC
#define NPROC      8
#endif

/* Process state constants */

#define PR_FREE      0      /* Process table entry is unused */
#define PR_CURR      1      /* Process is currently running */
#define PR_READY      2      /* Process is on ready queue */
#define PR_RECV      3      /* Process waiting for message */
#define PR_SLEEP      4      /* Process is sleeping */

/* Process state constants */

#define PR_FREE      0      /* Process table entry is unused */
#define PR_CURR      1      /* Process is currently running */
#define PR_READY      2      /* Process is on ready queue */
#define PR_RECV      3      /* Process waiting for message */
#define PR_SLEEP      4      /* Process is sleeping */
#define PR_SUSP      5      /* Process is suspended */
#define PR_WAIT      6      /* Process is on semaphore queue */
#define PR_RECTIM      7      /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN      16      /* Length of process "name" */
#define NULLPROC      0      /* ID of the null process */

/* Process initialization constants */

#define INITSTK      65536 /* Initial process stack size */
#define INITPRI      20    /* Initial process priority */
#define INITRET      userret /* Address to which process returns */

/* Inline code to check process ID (assumes interrupts are disabled) */

#define isbadpid(x)      ( ((pid32)(x) < 0) || \
    ((pid32)(x) >= NPROC) || \
    (proctab[(x)].prstate == PR_FREE) )
```

- Mengubah NPROC dari 8 ke 150

```

Activities Terminal Fri 1 May, 17:34
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
GNU nano 2.2.6 File: /home/xinu/xinu/include/process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */

umsg32 prmsg; /* Message sent to this process */
bool8 prhasmsg; /* Nonzero iff msg is valid */
int16 prdesc[NDESC]; /* Device descriptors for process */
};

/* Marker for the top of a process stack (used to help detect overflow) */
#define STACKMAGIC 0x0A0AAAA9

extern struct procent proctab[];
extern int32 prcount; /* Currently active processes */
extern pid32 currpids; /* Currently executing process */
xinu@xinu-develop-end:~$ nano /home/xinu/xinu/include/process.h
xinu@xinu-develop-end:~$ grep -n "NPROC" ~/xinu/include/process.h
5:#ifndef NPROC
6:#define NPROC 150
34: ((pid32)(x) >= NPROC) || \
xinu@xinu-develop-end:~$

```

2. Jalankan kode main_ex1!

```

/* main.c - main*/

#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    sndA();
    sndB();
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}

```



```

Activities Terminal Fri 1 May, 18:18
xinu@xinu-develop-end: ~/xinu/compile

Xinu for Vbox -- version #142 (xinu) Fri 1 May 21:16:02 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
  [0x0015E320 to 0x0009FFF7]
  [0x00100000 to 0x005FBFFF]
116457 bytes of Xinu code.
  [0x00100000 to 0x0011C6E8]
148360 bytes of data.
  [0x0011FF40 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
A
B
A
B
A
B

```

4. Jalankan kode main_ex3!

```

/* main.c - main*/

#include <xinu.h>
void sndChar(char);

process main(void)
{
    resume(create(sndChar, 1024, 20, "Send A", 1, 'A'));
    resume(create(sndChar, 1024, 20, "Send B", 1, 'B'));
    return OK;
}

void sndChar(char c)
{
    while(1){
        printf("%c \n", c);
        sleeps(1000);
    }
}

```

```

Activities Terminal Fri 1 May, 18:29
xinu@xinu-develop-end: ~/xinu/compile

Xinu for Vbox -- version #144 (xinu) Fri 1 May 21:27:10 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
  [0x0015E320 to 0x0009FFF7]
  [0x00100000 to 0x005FBFFF]
116425 bytes of Xinu code.
  [0x00100000 to 0x0011C6C8]
148360 bytes of data.
  [0x0011FF40 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
A
B
A
B
A
B

```

5. Buatlah 2 proses produser dan konsumen. Produser memproduksi angka integer dari 1-1000. Konsumer mengkonsumsi integer yang diproduksi oleh produser dan menampilkannya!

(Gunakan variabel global bertipe int32 bernama n yang digunakan secara bersama oleh kedua proses)

```
/* main.c - main*/

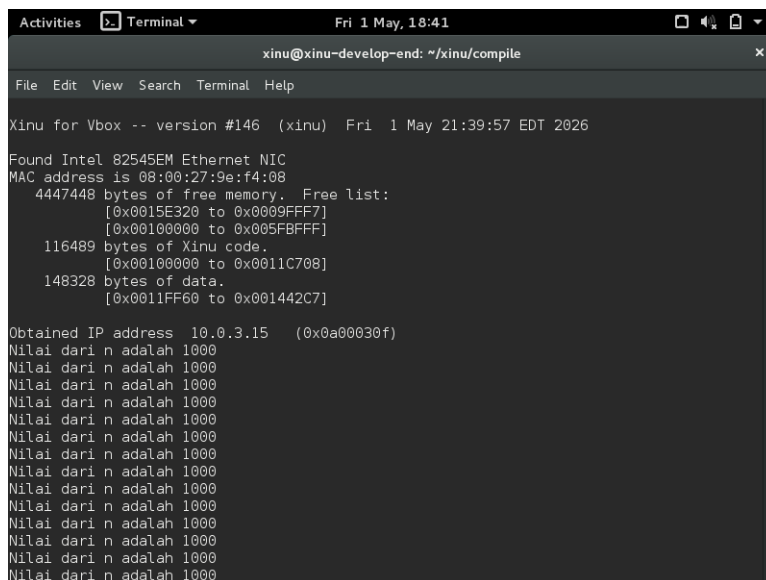
#include <xinu.h>

int32 n = 0;

void produser(void) {
    int32 i;
    for (i = 1; i <= 1000; i++) {
        n++;
    }
}

void konsumen(void) {
    int32 i;
    for (i = 1; i <= 1000; i++) {
        printf("Nilai dari n adalah %d\n", n);
    }
}

process main(void) {
    resume(create(produser, 1024, 20, "producer", 0));
    resume(create(konsumer, 1024, 20, "consumer", 0));
    return OK;
}
```



```
Activities Terminal Fri 1 May, 18:41
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
Xinu for Vbox -- version #146 (xinu) Fri 1 May 21:39:57 EDT 2026
Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
116489 bytes of Xinu code.
[0x00100000 to 0x0011C708]
148328 bytes of data.
[0x0011FF60 to 0x001442C7]
Obtained IP address 10.0.3.15 (0x0a00030f)
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
Nilai dari n adalah 1000
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Nilai dari n adalah 1000
Nilai dari n adalah 1000
```

Hasil program menunjukkan bahwa nilai n selalu 1000 karena proses produser menyelesaikan seluruh iterasi terlebih dahulu sebelum proses konsumen dijalankan. Hal ini terjadi karena tidak adanya delay atau mekanisme pembagian waktu yang memaksa kedua proses berjalan secara bergantian. Meskipun demikian, program ini tetap memiliki potensi race condition karena variabel n diakses secara bersama tanpa sinkronisasi. Jika penjadwalan proses berbeda, maka hasil yang ditampilkan dapat menjadi tidak teratur atau tidak konsisten.